THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · NCERT-TYPE FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A solution reacts with crushed egg-shells to give a gas that turns lime water milky. The solution contains: 1

(a) NaCl **(b)** HCl **(c)** Na_2SO_4 **(d)** NaOH

2. Which of the following methods is suitable for preventing an iron frying pan from rusting? 1

(a) applying grease **(b)** applying a coating of zinc
(c) applying paint **(d)** all of these

3. The number of covalent bonds in a molecule of pentane C_5H_{12} is: 1

(a) 15 **(b)** 14 **(c)** 17 **(d)** 16

4. The xylem in plants is responsible for: 1

(a) transport of amino acids **(b)** transport of oxygen
(c) transport of water **(d)** transport of food

5. The gap between two neurons is called: 1

(a) axon **(b)** myelin **(c)** dendrite **(d)** synapse

6. Multiple fission is seen in 1

(a) Amoeba **(b)** Hydra **(c)** Planaria **(d)** Plasmodium

7. Mendel chose the garden pea for his experiments because 1

(a) it grows only in cold regions
(b) it has no flowers
(c) it has many contrasting characters and is easy to grow
(d) it reproduces only asexually

8. The image formed by a plane mirror is 1

(a) real and diminished
(b) real and inverted
(c) virtual and magnified
(d) virtual, erect and of the same size as the object

9. Most of the refraction of light entering the eye takes place at the 1

(a) retina **(b)** outer surface of the cornea
(c) iris **(d)** eye lens

10. The SI unit of electric current is 1

(a) ampere **(b)** volt **(c)** coulomb **(d)** watt

11. Magnetic field lines inside a bar magnet go from 1

(a) the south pole to the north pole **(b)** the north pole to the south pole

(c) one side to the other randomly **(d)** do not exist

12. Which of the following is an abiotic component of an ecosystem? 1

(a) Grass **(b)** Temperature **(c)** Fungi **(d)** Bacteria

13. The white precipitate formed on mixing sodium sulphate and barium chloride solutions is: 1

(a) Na_2SO_4 **(b)** $NaCl$ **(c)** $BaSO_4$ **(d)** $BaCl_2$

14. Tooth decay starts when the pH of the mouth falls below: 1

(a) 3 **(b)** 7 **(c)** 9 **(d)** 5.5

15. A metal that reacts with cold water to produce hydrogen but floats because of bubbles of hydrogen: 1

(a) sodium **(b)** calcium **(c)** potassium **(d)** magnesium

16. The cleansing action of soap is due to: 1

(a) reaction with dirt **(b)** formation of micelles

(c) its smell **(d)** its acidity

17. In muscles during vigorous exercise, glucose breaks down anaerobically to give: 1

(a) water **(b)** ethanol **(c)** lactic acid **(d)** CO_2 only

18. The roots of a plant grow downward in response to: 1

(a) light **(b)** air **(c)** gravity **(d)** water only

19. The part of the flower that becomes the fruit is the 1

(a) stigma **(b)** ovule **(c)** style **(d)** ovary

20. The F2 phenotypic ratio in the dihybrid cross is 1

(a) 9 : 3 : 3 : 1 **(b)** 1 : 1 : 1 : 1 **(c)** 1 : 2 : 1 **(d)** 3 : 1

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Differentiate between biotic and abiotic components of an ecosystem. 2

22. Why do we apply paint on iron articles? 2

23. Why does an aqueous solution of an acid conduct electricity? 2

24. What is aqua regia? Why is it important? 2

25. What is a functional group? Name the one in ethanol. 2

26. How is the small intestine designed to absorb digested food? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Explain the oxidation of ethanol to ethanoic acid. Name two oxidising agents. 3

28. Describe the working of the nephron in forming urine. 3

29. Name three animal hormones, the glands and the functions. 3

30. Name the parts of a flower with their functions: 3
(a) anther
(b) stigma
(c) ovary.

31. Explain why the father, not the mother, determines the sex of the child. 3

OR

A convex lens has focal length 10 cm. An object is placed 30 cm from it. Find the image distance and magnification. 3

32. Why is it difficult for older persons to see nearby objects? 3

OR

Find the equivalent resistance when 2 ohm, 3 ohm and 5 ohm are connected in series, and the current from a 6 V battery. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Explain how auxin brings about phototropic bending of a shoot. 5

34. Describe the human female reproductive system. Where does fertilisation occur and where does the zygote develop? 5

35. Explain Mendel's experiments and the conclusions drawn from them. 5

36. Define power of a lens and state its unit. A convex lens of focal length 25 cm and a concave lens of focal length 50 cm are in contact. Find the net power. 5

OR

What is dispersion of light? Describe Newton's experiment with two prisms. 5

37. Establish the formula H = I^2 R t from the definitions of work and potential difference. 5

OR

Draw (describe) a domestic electric circuit and explain the functions of the live wire, neutral wire, earth wire and fuse. 5

38. What is biomagnification? Explain with an example of a food chain how a non-degradable pesticide gets concentrated. 5

OR

Describe what happens when a magnesium ribbon is burnt in air. Write the equation, name the type of reaction and state the nature of the product. 5

END OF QUESTION PAPER — Total: 80 Marks

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